Multi-Agent Distributed Lifelong Learning for Collective Knowledge Acquisition

Original File: 1709.05412v2.pdf | Generated on: June 10, 2026

1. Overall Summary

This paper introduces the Collective Lifelong Learning Algorithm (COLLA), addressing the limitations of traditional centralized lifelong learning in multi-agent, distributed environments. Current methods rely on a single agent with full data access, which is impractical for scenarios involving local data privacy, costly communication, or the need for parallel processing. COLLA enables a network of agents to collectively learn a series of tasks by formulating the problem as an online multi-task learning optimization. Agents learn local parametric models and share their knowledge bases with neighbors using an Alternating Direction Method of Multipliers (ADMM) algorithm, ensuring distributed computation and local data privacy. The approach provides theoretical guarantees for robust performance. Empirical evaluations on various benchmark datasets (Land Mine, Facial Expression, Computer Survey, London Schools) demonstrate that COLLA consistently outperforms existing distributed multi-task learning methods like ELLA, GO-MTL, and STL. The results show significant improvements in learning speed and asymptotic performance, with denser communication networks further enhancing learning rates. COLLA also provides a strong 'jumpstart' for new tasks, highlighting its effectiveness in knowledge transfer. This research offers a robust framework for collaborative knowledge acquisition in dynamic, decentralized multi-agent systems.

2. Main Idea

The central concept is to extend lifelong learning from a single agent to a network of multiple agents that collectively acquire knowledge over a series of tasks. The paper proposes the Collective Lifelong Learning Algorithm (COLLA), a distributed and decentralized approach that allows agents to share learned knowledge efficiently while preserving the privacy of their local data, thereby improving task performance and learning speed.

3. Problem Being Solved

This paper addresses several weaknesses and research gaps in previous lifelong learning systems. Primarily, it solves the problem that current lifelong learning algorithms rely on a single learning agent with centralized access to all data. This centralized paradigm is inefficient and impractical in real-world scenarios where data is distributed, local data privacy must be preserved, data communication is costly or time-consuming, and parallel processing is essential due to large data sizes or geographical distribution. The paper aims to enable multiple agents to collectively learn a series of tasks in a distributed and decentralized manner, overcoming the limitations of single-agent, centralized learning while facilitating knowledge sharing and accumulation.

4. Assumptions

- Agents in the network are synchronous, meaning they receive and learn one task simultaneously at each time step.
- There is a latent knowledge base that underlies all tasks, and each agent learns a local view of this knowledge base based on its own observations.
- The network graph is connected, allowing for knowledge and information flow between all agents.
- All nodes (agents) in the network are homogeneous, implying no central node guides collaboration.
- The data distribution for tasks has a compact support, which enforces boundedness on task-specific parameters and Hessian matrices.
- The LASSO problem (Eq. 4) for sparse coefficients admits a unique solution.
- The matrices formed by the Hessian of the loss function are strictly positive definite, ensuring that the functions are strongly convex.
- The arbitrary order of agents is known and fixed.

5. Limitations

- The paper does not address privacy considerations that may arise from transferring learned knowledge representations between agents, only preserving the privacy of locally observed raw data.
- The proposed framework assumes synchronous agents; extending it to a network of asynchronous agents with dynamic links is a future research direction.
- The current formulation requires an arbitrary but known and fixed order of agents, which can be restrictive.
- The inner loop of the algorithm requires a sufficiently large number of iterations (K) at each time step to ensure agents converge to an agreement, which can impact computational cost.

6. Results and Conclusion

The experimental findings demonstrate that the Collective Lifelong Learning Algorithm (COLLA) significantly outperforms existing distributed multi-task learning approaches, including ELLA, GO-MTL, and single-task learning (STL), across various classification and regression benchmark datasets (Land Mine, Facial Expression, Computer Survey, London Schools). Collaboration among agents, facilitated by COLLA, leads to improved lifelong learning performance, both in terms of faster learning speed and higher asymptotic accuracy, reaching levels unattainable by a single agent. The performance of the online COLLA algorithm is comparable to or even surpasses offline distributed multi-task learning methods, validating its effectiveness in dynamic settings. COLLA also exhibits a strong 'jumpstart' capability, indicating that knowledge transferred from previous tasks significantly boosts initial performance on new tasks. Moreover, the research concludes that network topologies with more edges, allowing for greater communication and collaboration, result in faster learning rates and better overall performance.

7. Real World Impact

This research has practical applications in various real-world domains. It can be applied to personal intelligent robot assistants and chatbots that continuously interact with different users, allowing them to adapt and improve over time. Financial decision support agents can benefit by collectively learning from partially accessible multi-modal data views. In healthcare, it enables collaborative learning while adhering to regulations that prevent sharing of personal medical data. Home service robots and wearable devices can process perceptions locally, reducing communication costs and bandwidth limitations. Furthermore, the approach is suitable for modern big data systems that require parallel processing in the cloud across multiple virtual agents (CPUs or GPUs), and generally for autonomous agents and multiagent systems requiring collective knowledge acquisition.

8. Graphs and Figures Analysis

Figure 1: Performance of distributed (dotted lines), centralized (solid), and single-task learning (dashed) algorithms on benchmark datasets. Shaded region shows standard error (Best viewed in color.)

This figure presents the performance of different learning algorithms (COLLA, ELLA, Dist. CoLLA, GO-MTL, STL) on four benchmark datasets, measured by AUC for classification tasks and RMSE for regression tasks. The x-axis represents the number of tasks learned, and the y-axis shows the performance metric. Shaded regions indicate standard error. Across all datasets, COLLA (dotted blue) and Dist. CoLLA (dotted green) consistently outperform ELLA (solid red), GO-MTL (solid purple), and STL (dashed black). For classification tasks (Land Mine and Facial Expression), COLLA and Dist. CoLLA achieve higher AUC values faster and maintain superior asymptotic performance. For regression tasks (Computer Survey and London Schools), they achieve lower RMSE values more quickly and maintain better performance. This demonstrates that collective and distributed lifelong learning, as implemented by COLLA, significantly improves both learning speed and final performance compared to single-agent or centralized multi-task learning approaches.

Figure 2: Performance of CoLLA given various graph structures (a) for three data sets (b-d).

This figure illustrates the impact of different network graph structures on the performance of the CoLLA algorithm across three datasets. Figure 2a visually depicts four graph topologies: Random, Complete, Tree, and Server. Figures 2b-2d show the performance (AUC for Facial Expression, RMSE for Computer Survey and London Schools) as a function of the number of tasks for each graph structure. For Facial Expression (2b), Complete and Server structures yield higher AUC, indicating better classification performance. Similarly, for Computer Survey (2c) and London Schools (2d), Complete and Server structures result in lower RMSE, signifying improved regression performance. The trend across these plots indicates that network structures with more edges and thus greater communication and collaboration (like Complete and Server graphs) lead to faster learning rates and superior overall performance compared to sparser structures (like Random and Tree graphs).

Table 1: Jumpstart comparison (improvement in percentage) on the Land Mine (LM), London Schools (LS), Computer Survey(CS), and Facial Expression (FE) datasets

This table compares the 'jumpstart' performance, which is the initial performance improvement on a new task due to knowledge transfer, measured in percentage, for different algorithms on four datasets: Land Mine (LM), London Schools (LS), Computer Survey (CS), and Facial Expression (FE). COLLA shows notable jumpstart improvements across all datasets (e.g., 51.44% on CS, 40.87% on FE). ELLA also shows improvements but generally less than COLLA. Distributed COLLA (offline) often achieves the highest jumpstart percentages (e.g., 61.71% on CS, 59.89% on FE), followed closely by GO-MTL (offline centralized). This indicates that COLLA effectively leverages transferred knowledge to provide a significant initial performance boost on new tasks, with offline distributed methods showing even stronger initial gains due to simultaneous access to all tasks.